

EXPERIMENT 6

6 Natural Cooling of Hot Water: Plotting Temperature-Time Graph

Aim

To measure the temperature of hot water during natural cooling and plot the temperature versus time graph.

Requirements

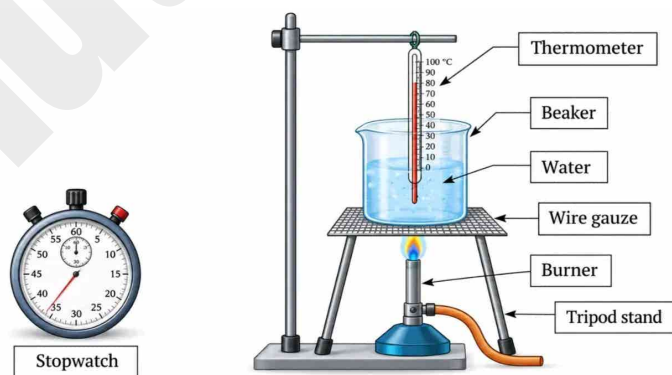
A 250 ml glass beaker, a thermometer (0°C – 100°C), a stopwatch (or a clock with hour, minute and second hands), a stand, a tripod, a burner, a wire gauze, and water.

Procedure

1. Determine the least count of the thermometer.
2. Pour 150 ml of water into the beaker.
3. Set up the apparatus as shown in the figure, making sure the bulb of the thermometer stays about 2 cm above the bottom of the beaker.
4. Heat the water.
5. Switch off the burner once the water reaches a temperature of 80°C .
6. When the temperature drops to 70°C , start the stopwatch and note the temperature of the water every minute.
7. Keep noting the temperature until the water reaches room temperature.
8. Plot the graph of temperature versus time.

Diagram

(Label the various components in the given diagram)



Natural cooling of hot water

Observations

1. Least count of the thermometer = 1°C .
2. Initial temperature of water = 70°C .

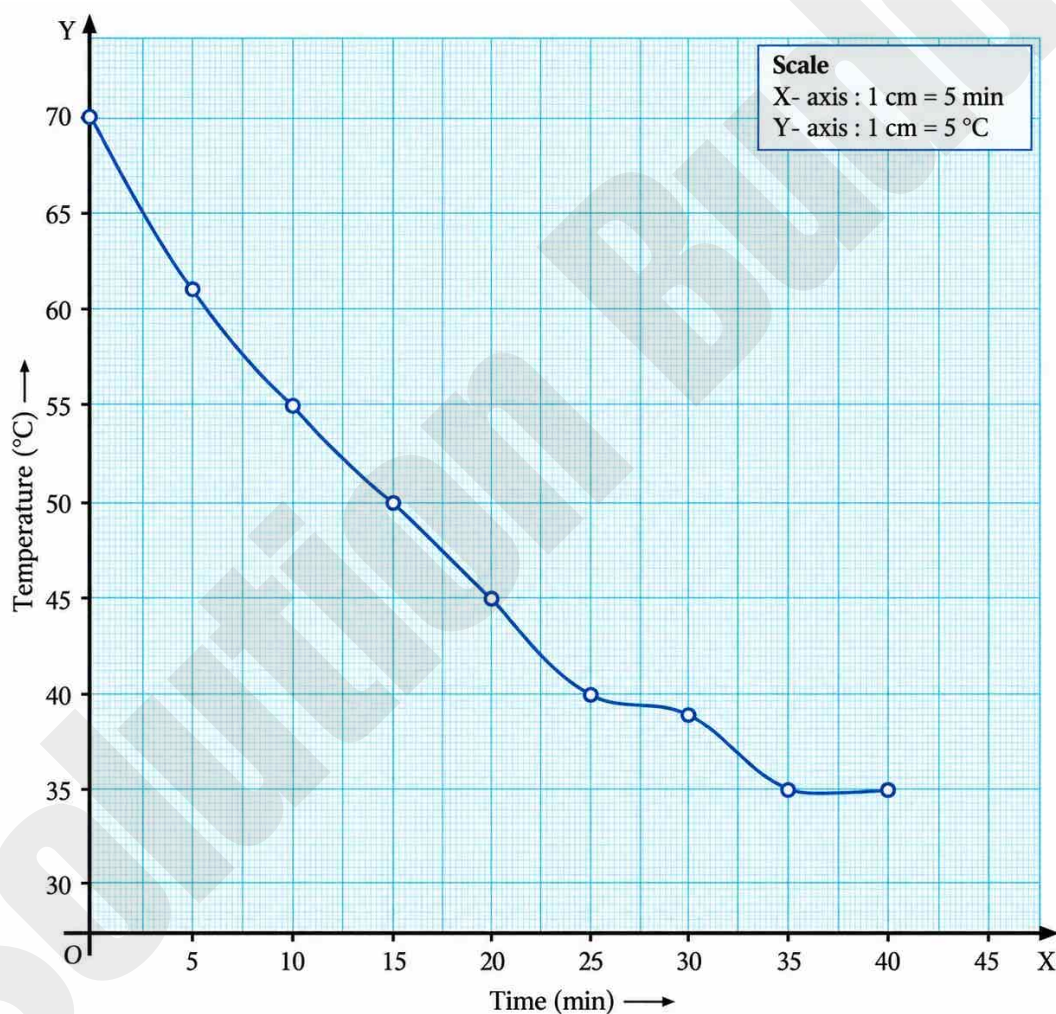
Observation Table

Time (min)	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Temperature ($^{\circ}\text{C}$)	70	68	66	64	62	61	60	59	58	56	54	53	51	49

Time (min)	14	15	16	17	18	19	20	21	22	23	24	25	26	27
Temperature ($^{\circ}\text{C}$)	51	50	49		46		44					40		

Time (min)	28	29	30	31	32	33	34	35	36	37	38	39	40
Temperature ($^{\circ}\text{C}$)			38					35					34

Graph of Temperature versus Time



Graph of temperature versus time

Scale — X-axis: 1 cm = 5 min | Y-axis: 1 cm = 5°C

Inferences / Conclusions

1. The hot water loses heat to the surroundings. This results in lowering of the temperature of the water.
2. The rate of cooling depends on the excess of temperature of water over the surroundings. When the difference between the temperature of hot water and the temperature of the surroundings is more, the rate of cooling is more.

Precautions

1. Switch off the ceiling fans during the experiment.
2. The bulb of the thermometer should not touch the bottom of the beaker.

Multiple Choice Questions

1. Which of the following is the correct relation?

- (A) 4.18 calories = 1 joule (B) 1 calorie = 4.18 joules
(C) 10^7 calories = 1 joule (D) 1 calorie = 10^7 joule

Answer: (B) 1 calorie = 4.18 joules

2. Which of the following are the processes of transfer of heat?

- (A) conduction (B) convection
(C) radiation (D) All the above

Answer: (D) All the above

3. The temperature at which water changes into steam is called the of water.

- (A) melting point (B) boiling point (C) expansion point (D) dew point

Answer: (B) boiling point

4. Heat is absorbed when

- (A) water is converted into vapour. (B) water is converted into ice.
(C) steam is converted into water. (D) all of these

Answer: (A) water is converted into vapour.

Date: _____

Teacher's Signature: _____